

Masked Isolated Sharded Distillation: A Hybrid Machine Unlearning Approach for Multimodal Data

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Abstract—As the training of artificial intelligence models on data sourced from various media platforms keeps increasing, data privacy has become an important issue. Data privacy regulations such as the General Data Protection Regulation (GDPR) started to dictate the right to be forgotten. This conflict resulted in the concept of machine unlearning as a critical area of research. However, the existing machine unlearning approaches are commonly evaluated on image classification tasks without revealing their applicability or effectiveness in other media modalities. In this paper, we propose Masked Isolated Sharded disTillation (MIST), a novel class-based machine unlearning method that utilizes sharded training with masked knowledge distillation. Our method quickly and efficiently erases the effect of the selected class while preserving the knowledge of remaining data. We conduct the experiments across six datasets including image, audio and video modalities. Furthermore, we benchmark our method with nine well-known unlearning methods in the literature. Tests are evaluated through forget accuracy, retain accuracy, H-Mean, privacy and computational cost metrics. These metrics confirm that the proposed hybrid method achieves highly effective unlearning strength with strong privacy performance and consistency on cross-modal platforms.

Index Terms—machine unlearning, multimedia, data privacy, sharded training, knowledge distillation.

I. INTRODUCTION

In recent years, with the increasing usage of various machine learning models such as Large Language Models (LLMs), a vast amount of personal and private data has started to be used to train those models. After the training, that personal data become embedded in the model’s weights, creating a critical vulnerability about data privacy. Removing the source data from the training dataset does not solve that vulnerability, still leaving the effect of that data on the model’s weights. Regulations like the General Data Protection Regulation (GDPR) [17] now demand the deletion of this encoded influence upon personal request [1]–[4]. Furthermore, the training data used by machine learning models can wear out, leading to unwanted behaviors and mistakes in different fields of machine learning. For example hallucinations happening in LLMs provide misleading and incorrect responses even to the simplest prompt of the user. These stated problems give rise to the concept of machine unlearning, aiming to remove the impact of the selected data.

In the literature, the image-only focus leaves a critical gap regarding the implementation of these methods on other modalities. Modern machine learning systems operate on various media data, including video streams from surveillance and

recordings from voice assistants and call centers which have their own risks and vulnerabilities of personal data privacy. A recent survey on multimodal unlearning states that machine unlearning on video and audio modalities remains largely unexplored compared to machine unlearning on image and language modalities [13].

In an effort to bridge this gap, we propose MIST (Masked Isolated Sharded disTillation), a hybrid machine unlearning framework that combines two complementary and powerful machine unlearning concepts. One is SISA-based data partitioning which provides structural isolation by splitting training data into independent shards, and the other is the knowledge distillation with logit masking inspired by DELETE [6]. By utilizing these two techniques, MIST allows the elimination of the influence of a selected class within each affected shard. MIST provides scalability via sharding the training data, and powerful unlearning performance by localizing the knowledge distillation in each shard, minimizing the noise generated by other data samples. To the best of our knowledge, MIST is the first machine unlearning approach handling multiple modalities.

The main contributions of this work are as follows:

- We introduce MIST, a machine unlearning architecture that utilizes SISA’s sharding approach with DELETE’s masked distillation mechanism. This method enables efficient class-level unlearning with accuracy preservation on the remaining data across different data modalities.
- We propose a comprehensive and transparent benchmark framework using four image classification datasets. This framework is comparing MIST with nine well-known unlearning methods with various metrics and Membership Inference Attack (MIA) [15] privacy analysis. These metrics consist of the ones widely used in the literature and a newly defined one, Remain Accuracy Preservation (RAP) ratio. RAP ratio displays the change of retain accuracy rates of the baseline model after the unlearning.
- In order to investigate the efficiency of MIST further, we conducted class-sweep unlearning tests in 6 datasets including image, audio and video modalities. By unlearning each class in the datasets and providing a standard deviation and average values of some important metrics, we revealed that MIST can operate in these datasets regardless of which class we choose.

II. RELATED WORK

A. Exact Unlearning

Machine unlearning can generally be divided into two categories: exact unlearning and approximate unlearning [1]–[4]. Exact unlearning methods aim to remove the influence of the target data from the model’s parameters completely. The simplest approach is retraining from scratch after removing the target samples. This method remains the gold-standard baseline because this method produces a model that has never seen the target data. [9]. However, it requires repeating the full training process for every deletion request in related data, which is inefficient for the time and computational cost [3].

SISA (Sharded, Isolated, Sliced, and Aggregated) [5] is an alternative solution in the exact unlearning concept. This method divides the entire training dataset into isolated shards. For each unlearning request, only the relevant shards need to be retrained. The remaining shards do not need to be retrained; because of that, they will not be affected by the deletion request. However, handling the checkpoints in the multi-shard systems might be challenging on larger datasets [5].

B. Approximate Unlearning

Approximate unlearning algorithms update the parameters of a previously trained model without retraining. Because of that, they are faster than exact unlearning methods but these methods cannot guarantee that the target/forget class is completely forgotten [2], [3].

DELETE employs masked knowledge distillation for its unlearning procedure [6]. In this procedure, it uses the original training model as the teacher model, sets the logit value of the target class to a large negative number, and finally updates a student model that matches this masked logit. This approach leads to reducing the probability value of the target/forget class while it is preserving information in remaining classes. Amnesiac [7] reduces the training updates on target samples by using per-sample gradient values to reverse the updates. Boundary Unlearning [8] pushed the decision boundary away from the target class without updating weights of the model.

Recent works combine both approaches into a single methodology. Li et al. [10] use both partial retraining and direct parameter update for unlearning. Quan et al. [11] propose verified unlearning, which offers guaranteed unlearning mechanisms in distillation-based scenarios.

C. Multimodal Unlearning

Most of the machine unlearning studies have been focused on image-based datasets. The recent survey by Sarwar et al. [13] discusses various types of modalities’ unlearning such as vision, language, video, and audio. As highlighted in the survey, machine unlearning research in video and audio has not received enough attention yet. For audio task, Mason-Williams et al. [16] are the first researchers who provide a comprehensive review on unlearning in the audio domain. This study confirms that the modality gap between image and audio unlearning is real and largely unaddressed. In the video modality concept, although 3D convolutional neural networks

are widely used for action recognition [14], there is no study that focuses on unlearning for this type of data.

III. PROPOSED UNLEARNING METHOD: MIST

A. Overview

The proposed MIST framework, illustrated in Fig. 1, consists of three stages that perform different operations. In the *Sharded Training* stage, the training dataset is divided into K separate shards, each used to train separate and independent sub-models. When a class-level unlearning request is received, the *Masked Distillation* stage applies a teacher-student knowledge distillation procedure with logit masking to each shard model. In order to test and use the unlearned shard models, the input data gets processed over each unlearned shard model. After the unlearning process, outputs of those models are fed into the *Entropy Based Aggregation* stage. In this stage, the outputs of each shard model are being prioritized by their entropy values and get aggregated to provide a single output.

B. Sharding Module

In this module, the training dataset $\mathcal{D}$ with N samples is partitioned into K disjoint shards $\{S_1, S_2, \dots, S_K\}$ such that:

$$\bigcup_{i=1}^K S_i = \mathcal{D}, \quad S_i \cap S_j = \emptyset \quad \forall i \neq j, \quad |S_i| \approx \frac{N}{K} \quad (1)$$

To ensure the balance of the class samples in each shard, all dataset indexes are first shuffled with a fixed seed and then evenly distributed [5]. The next step is that each shard S_i is employed to train the corresponding individual model M_i . This enables us to perform an unlearning operation by manipulating those models related to the requested data only.

C. Masked Distillation Phase

When a forget request for class c_f is received, this phase is applied to each shard. For each shard i , two copies are created: a teacher model with frozen weights and a student model. The teacher’s logit for the forget class is then masked to a large negative number, -10^{10} in our case. This mask forces the softmax outputs to redistribute probability mass to non-forget classes. Then, for the student to learn this masked distribution, the KL-divergence loss [26] is applied between the student and the teacher models. At the same time, a cross-entropy term on remaining data prevents the potential damage to the remaining class knowledge. The hyperparameter λ shown in Algorithm 1 controls the balance between unlearning strength and the knowledge preservation of the remaining set. Another technique to optimize the unlearning process is gradient clipping with norm γ [25], which limits the maximum gradient update. This parameter prevents destabilized and forced weight updates from damaging the remaining dataset knowledge.

D. Entropy-Weighted Aggregation

After unlearning, some of the shards can get harmed more than others regarding their retain accuracies after the unlearning process. To sum up all the shard outputs, shard predictions are combined via entropy-weighted averaging as shown in

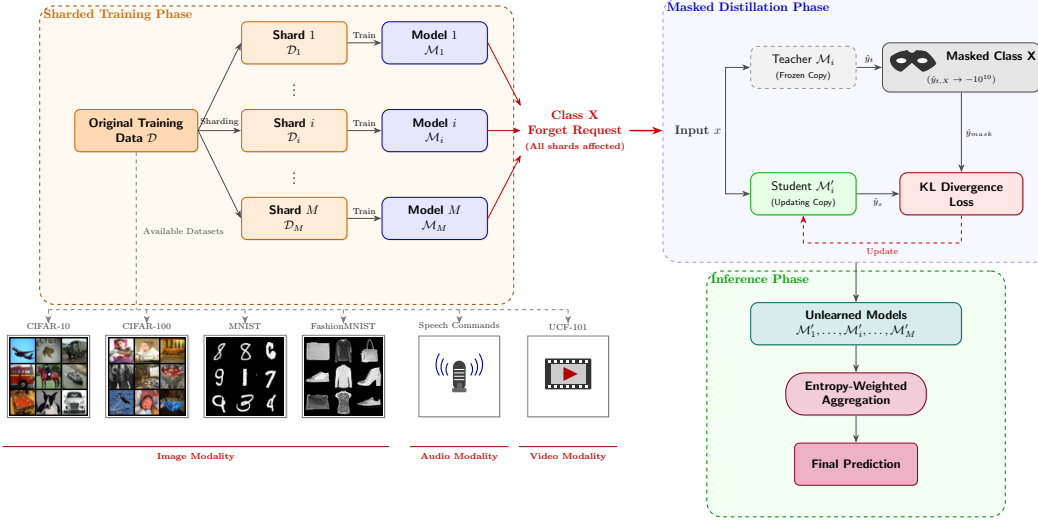


Fig. 1. Overview of the proposed MIST framework with three phases: Sharded Training, Masked Distillation, and Entropy-Weighted Inference Aggregation.

Algorithm 1 Masked Distillation for Shard Unlearning

Input: Forget class c_f , Shard model $\mathcal{M}_i$, Forget set $\mathcal{D}_f^i$, Remain set $\mathcal{D}_r^i$, Epochs E , Learning rate η , Remain regularization weight λ , Clip norm γ

Output: Unlearned model $\mathcal{M}'_i$

- 1: $\mathcal{M}'_i \leftarrow \mathcal{M}_i$ ▷ Initialize student
- 2: Freeze $\mathcal{M}_i$ as teacher
- 3: **for** epoch = 1 **to** E **do**
- 4: **for each** batch (x_f, y_f) **in** $\mathcal{D}_f^i$ **do**
- 5: $\hat{z}_T \leftarrow \mathcal{M}_i(x_f)$ ▷ Teacher logits
- 6: $\hat{z}_T[c_f] \leftarrow -10^{10}$ ▷ Mask forget class
- 7: $\hat{z}_S \leftarrow \mathcal{M}'_i(x_f)$ ▷ Student logits
- 8: $\mathcal{L}_{\text{forget}} \leftarrow D_{\text{KL}}(\text{Softmax}(\hat{z}_T) \parallel \text{Softmax}(\hat{z}_S))$
- 9: Sample batch (x_r, y_r) from $\mathcal{D}_r^i$ ▷ Remain reg.
- 10: $\mathcal{L}_{\text{remain}} \leftarrow \text{CE}(\mathcal{M}'_i(x_r), y_r)$
- 11: $\mathcal{L} \leftarrow \mathcal{L}_{\text{forget}} + \lambda \cdot \mathcal{L}_{\text{remain}}$
- 12: Clip $\nabla \mathcal{L}$ to max norm γ
- 13: Update $\mathcal{M}'_i$ with $\eta \cdot \nabla \mathcal{L}$
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathcal{M}'_i$

Algorithm 2. Each shard model produces a probability distribution, and its prediction entropy determines the aggregation weight. This weight allows that more confident shards (lower entropy) receive higher weight. This naturally lowers the effect of the shards that have high entropy after unlearning.

E. Cross-Modal Adaptability

The unlearning phase of MIST masks the forget class prediction scores on the output logit space, making modality differences irrelevant. It is also observed that the sharding and aggregation phases are agnostic to data format. MIST can be applied in machine learning tasks that share a common output format: a logit vector that has a dimension of the number

of classes $z \in \mathbb{R}^C$, where the elements correspond to the prediction score of each class. Three core stages of MIST remain structurally identical across multiple modalities. Only the backbone architecture, data preprocessing, and unlearning hyperparameters are updated in each modality. For the audio keyword spotting task, an optional fine-tuning mechanism is added on the remaining data. This mechanism applies cross-entropy on the whole remaining set after masked distillation. This additional step normalizes model predictions on the remaining class, reducing the detectability of the unlearning operations.

Algorithm 2 Entropy-Weighted Aggregation

Input: Test sample x , Unlearned shard models $\{\mathcal{M}'_1, \dots, \mathcal{M}'_K\}$, Temperature T

Output: Final prediction y_{final}

- 1: **for** $i = 1$ **to** K **do**
- 2: $p_i \leftarrow \text{Softmax}(\mathcal{M}'_i(x)/T)$ ▷ Scaled probabilities
- 3: $H_i \leftarrow -\sum_c p_{i,c} \log(p_{i,c})$ ▷ Entropy of shard i
- 4: $w_i \leftarrow \exp(-H_i)$ ▷ Lower entropy = higher weight
- 5: **end for**
- 6: $W \leftarrow \sum_{i=1}^K w_i$ ▷ Normalization factor
- 7: $p_{\text{agg}} \leftarrow \sum_{i=1}^K \frac{w_i}{W} p_i$ ▷ Weighted average
- 8: $y_{\text{final}} \leftarrow \arg \max(p_{\text{agg}})$
- 9: **return** y_{final}

IV. EXPERIMENTS

A. Experimental Setup

Experiments are divided into two phases. The first phase is method-level comparison that four image datasets, named CIFAR-10 [18], MNIST [19], CIFAR-100 [18], and Fashion-MNIST [20], are used. In the first phase, ResNet-18 [23] is used as a backbone model for all the unlearning methods because it provides as an important benchmark in image

based experiments. Also, all the methods are compared with the same dataset, the same forget class, same train/test split and same evaluation procedure. For the training procedure, sharding structure is used only for SISA and MIST, while other methods were evaluated according to common architecture. This setup provides a fair and consistent environment for comparing different unlearning methods.

In the second phase, MIST's class-sweep performance was evaluated across different modalities and classes. For image datasets, ResNet-50 [23] is employed to determine the behavior of MIST under strong feature representations. For audio and video modalities, modality-specific models are used because 2D image backbone models are not suitable for this type of datasets. We used audio CNN [24] model for Speech Commands [21] and 3D ResNet-18 [14] for UCF101 [22].

B. Evaluation Metrics

In this study, various metrics are used to evaluate unlearning models' performance. Acc_f represents the accuracy rate on the forget class, while Acc_r represents the average accuracy of remaining classes, excluding forget class. In Table I, the metrics Acc_f and Acc_r are used to evaluate forgetting and retention performance of the unlearning models, while Acc_{ft} and Acc_{rt} are used for testing split. Low Acc_f and Acc_{ft} values show stronger forgetting, while high Acc_r and Acc_{rt} values show that the remaining knowledge is preserved.

Also the $F1_{rt}$ metric measures the retain-side classification performance on the test set. RAP_{tr} and RAP_{te} show the retain accuracy preservation ratio for training and test splits respectively. RAP is defined as $Accr^u / Accr^b \times 100$, where u and b denote the unlearned and baseline pretrained models. H-Mean (Harmonic Mean) is used to measure the balance between the forgetting and retaining performance. Lastly, MIA (Membership Inference Attack) and time are used to calculate privacy risk and efficiency. Lower MIA and time metrics show a safer and more practical unlearning process. In Table I, the Time metric shows the unlearning phase for unlearning methods, while it represents the initial training cost in pretrained model rows.

C. Method-Level Comparison

Table I represents the method-level comparisons between MIST, exact and approximate unlearning approaches. Note that this table should not be evaluated solely on the forget accuracy. An ideal unlearning method should reduce both Acc_f and Acc_{ft} and keep a high level of Acc_r , Acc_{rt} , $F1_{rt}$, RAP_{tr} , and RAP_{te} metrics. Forget class is class 4 in each dataset. For these reasons, Table I includes comparisons between forgetting and retain-side performance, privacy, and unlearning time.

Unlearning is confirmed since the Acc_{ft} of MIST gets reduced to 0.00% for all four datasets, which means that the chosen forget class is successfully removed from the prediction mechanism of the machine learning model. In addition, MIST successfully retains the performance of the remaining classes through high RAP_{tr} and RAP_{te} scores, obtaining good H-Mean results in return. As mentioned above, the importance

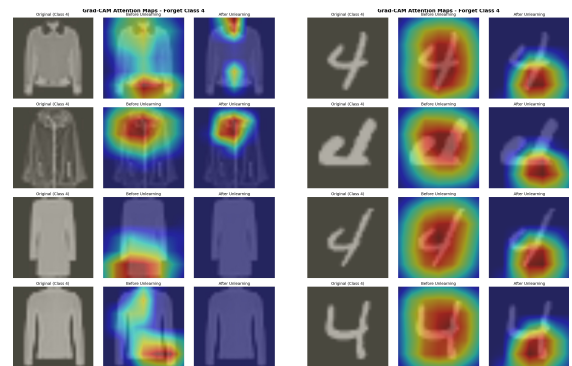
of high retain-side performance becomes crucial in CIFAR-10 and CIFAR-100, because the images in these datasets are much harder to recognize than those in MNIST datasets. Although some baselines provide similar H-Mean results to MIST on MNIST and FashionMNIST, MIST maintains its advantage in terms of complete forgetting, low MIA, good retain-side performance, and lower computational cost.

D. Class-Sweep and Cross-Modal Generalization

The class-sweep results of MIST across image, audio, and video datasets are shown in Table II. Unlike the previous method-based comparisons, this part evaluates the application of MIST outside image-based datasets. For this purpose, CIFAR-10, MNIST, FashionMNIST, and Speech Commands are analyzed using an average of 10 classes' results, while CIFAR-100 uses 100 classes, and UCF101 uses 101 classes. On image datasets, particularly in FashionMNIST and CIFAR-100; MIST shows good performance by getting low forget class accuracy and high remaining class accuracy. In addition, the Speech Commands results prove that MIST can be applied to audio data by showing good forget and retain accuracy. The video action recognition dataset, UCF101 is more difficult to unlearn because of the temporal motion and visual similarity of the video classes. However, if correct hyperparameter optimizations are applied into each class, unlearning can still be performed efficiently.

E. Privacy and Visual Analysis

In terms of privacy, the Membership Inference Attack (MIA) is used as the attack method for calculating if the unlearned model still contains any information about the forget samples [15]. In our case, threshold-based MIA is applied. The MIA score is determined based on three types of signals: correctness, confidence, and modified entropy. The lower MIA value implies that the attacker possesses little information to distinguish forget training samples from non-members. According to Table I, MIST achieves 0.00% MIA in both MNIST and FashionMNIST datasets. Because of that, it can be stated that the forget class does not contain any membership signal after the unlearning process. It should be considered that MIA alone cannot guarantee the complete forgetting; it must be supported with other evaluation metrics.



(a) FashionMNIST

(b) MNIST

Fig. 2. Grad-CAM attention maps for class 4 before and after unlearning.

TABLE I
COMPREHENSIVE METHOD COMPARISON BENCHMARK ON IMAGE DATASETS.

Dataset	Method	Train Accuracy		Test Accuracy		$F1_{rt} \uparrow$	$RAP_{tr} \uparrow$	$RAP_{te} \uparrow$	H-Mean $\uparrow$	MIA $\downarrow$	Time (s)
		$Acc_f \downarrow$	$Acc_r \uparrow$	$Acc_{ft} \downarrow$	$Acc_{rt} \uparrow$						
CIFAR-10	Pretrained Model	100.00%	100.00%	96.60%	94.72%	–	–	–	–	–	2520
	Pretrained Model (Sharded)	98.62%	98.09%	95.90%	94.97%	–	–	–	–	–	10764.0
	Retrain from Scratch	0.00%	100.00%	0.00%	94.62%	94.63%	100.00%	99.89%	97.24	0.00%	3925.2
	SISA	0.00%	96.16%	0.00%	93.24%	93.23%	98.03%	98.19%	96.50	0.00%	9405.0
	DELETE	0.00%	100.00%	0.00%	94.98%	94.99%	100.00%	100.27%	97.42	0.00%	626.6
	Fine-Tuning	0.00%	98.98%	0.00%	93.03%	93.04%	98.98%	98.22%	96.39	0.00%	269.5
	Gradient Ascent [27]	0.12%	57.57%	0.10%	53.32%	54.04%	57.57%	56.29%	69.53	0.73%	611.3
	SCRUB [28]	0.00%	39.65%	0.00%	41.88%	38.30%	39.65%	44.21%	59.03	0.00%	98.2
	AmnesiacML	0.00%	99.08%	0.00%	92.74%	92.76%	99.08%	97.91%	96.24	0.00%	166.3
	Boundary Shrink	0.12%	55.43%	1.10%	56.62%	56.69%	55.43%	59.78%	72.01	5.32%	666.3
	Boundary Expanding	16.96%	58.93%	2.20%	47.70%	48.23%	58.93%	50.36%	64.12	26.95%	610.5
	MIST (Ours)	0.00%	97.81%	0.00%	95.18%	95.18%	99.60%	100.22%	97.53	0.00%	160.1
MNIST	Pretrained Model	99.91%	99.82%	99.29%	99.10%	–	–	–	–	–	840.6
	Pretrained Model (Sharded)	99.49%	99.44%	99.08%	99.17%	–	–	–	–	–	1371.6
	Retrain from Scratch	0.00%	99.80%	0.00%	99.27%	99.27%	99.98%	100.17%	99.63	0.00%	1456.5
	SISA	0.00%	99.44%	0.00%	99.27%	99.27%	100.00%	100.10%	99.63	0.00%	1121.8
	DELETE	0.00%	99.80%	0.00%	99.16%	99.16%	99.97%	100.06%	99.58	0.00%	583.8
	Fine-Tuning	0.00%	99.93%	0.00%	99.38%	99.38%	100.11%	100.28%	99.69	0.00%	242.7
	Gradient Ascent	0.00%	10.94%	0.00%	10.87%	2.13%	10.96%	10.97%	19.60	0.00%	127.2
	SCRUB	0.00%	97.84%	0.00%	97.74%	97.73%	98.01%	98.62%	98.86	0.00%	99.0
	AmnesiacML	0.00%	100.00%	0.00%	99.60%	99.60%	100.18%	100.50%	99.80	0.00%	143.7
	Boundary Shrink	8.83%	99.28%	10.18%	98.46%	98.46%	99.46%	99.35%	93.94	3.17%	60.0
	Boundary Expanding	2.38%	94.96%	2.14%	94.72%	94.71%	95.13%	95.58%	96.27	0.79%	565.9
	MIST (Ours)	0.00%	99.34%	0.00%	99.08%	99.08%	99.89%	99.91%	99.54	0.00%	182.2
CIFAR-100	Pretrained Model	100.00%	99.97%	68.00%	82.97%	–	–	–	–	–	3960
	Pretrained Model (Sharded)	94.00%	96.03%	75.00%	80.47%	–	–	–	–	–	5503.3
	Retrain from Scratch	0.00%	99.81%	0.00%	79.27%	79.24%	99.84%	95.54%	88.44	0.00%	4390.4
	SISA	0.00%	90.86%	0.00%	80.36%	80.25%	94.62%	99.86%	89.11	0.00%	3514.4
	DELETE	1.40%	99.80%	1.00%	79.88%	79.73%	99.83%	96.27%	88.42	0.87%	2456.3
	Fine-Tuning	0.00%	98.79%	0.00%	79.80%	79.99%	98.82%	96.18%	88.76	0.00%	1236.7
	Gradient Ascent	0.80%	98.96%	0.00%	77.26%	76.95%	98.99%	93.12%	87.17	0.20%	2371.9
	SCRUB	1.20%	99.95%	0.00%	82.87%	82.85%	99.98%	99.88%	90.63	0.40%	329.0
	AmnesiacML	1.60%	98.91%	0.00%	79.75%	79.80%	98.94%	96.12%	88.73	0.33%	541.8
	Boundary Shrink	2.60%	98.29%	1.00%	74.58%	75.82%	98.32%	89.88%	85.07	1.20%	2494.3
	Boundary Expanding	33.60%	86.47%	25.00%	63.20%	64.60%	86.50%	76.17%	68.60	5.20%	2481.1
	MIST (Ours)	0.02%	95.81%	0.00%	79.66%	79.6%	99.78%	98.98%	88.68	0.67%	300.2
FashionMNIST	Pretrained Model	100.00%	100.00%	94.00%	95.33%	–	–	–	–	–	1694.4
	Pretrained Model (Sharded)	95.83%	96.96%	92.50%	94.13%	–	–	–	–	–	2565.6
	Retrain from Scratch	0.00%	98.25%	0.00%	92.11%	91.98%	98.25%	96.62%	95.89	0.00%	1957.7
	SISA	0.00%	97.23%	0.00%	94.96%	94.93%	100.28%	100.87%	97.41	0.00%	2162.5
	DELETE	0.00%	100.00%	0.00%	95.66%	95.63%	100.00%	100.34%	97.78	0.00%	831.6
	Fine-Tuning	0.00%	98.78%	0.00%	94.01%	93.98%	98.78%	98.61%	96.91	0.00%	332.1
	Gradient Ascent	0.00%	11.11%	0.00%	11.11%	2.22%	11.11%	11.66%	20.00	0.00%	183.9
	SCRUB	0.00%	29.53%	0.00%	29.44%	16.01%	29.53%	30.89%	45.49	0.00%	113.7
	AmnesiacML	0.00%	99.60%	0.00%	95.38%	95.38%	99.60%	100.05%	97.63	0.00%	190.9
	Boundary Shrink	43.63%	95.69%	38.50%	89.94%	90.02%	95.69%	94.35%	73.05	30.93%	72.2
	Boundary Expanding	0.28%	99.72%	0.40%	94.19%	94.11%	99.72%	98.80%	96.82	0.09%	783.9
	MIST (Ours)	0.00%	97.44%	0.00%	94.89%	94.85%	100.49%	100.80%	97.38	0.00%	156.4

TABLE II
CLASS-SWEEP GENERALIZATION RESULTS OF MIST ACROSS DIFFERENT MODALITIES.

Dataset	Modality	#Classes	Acc_f (%) $\downarrow$		Acc_{r-t} (%) $\uparrow$		RAP_{te} (%) $\uparrow$		H-Mean $\uparrow$		Unlearning Time (s)	
			mean $\pm$ std	[min, max]	mean $\pm$ std	[min, max]	mean $\pm$ std	[min, max]	mean $\pm$ std	[min, max]	mean $\pm$ std	[min, max]
CIFAR-10	Image	10	1.33 $\pm$ 1.24	[0.10, 4.50]	86.84 $\pm$ 1.37	[84.91, 89.78]	99.39 $\pm$ 0.89	[98.06, 101.34]	92.37 $\pm$ 0.67	[91.67, 94.07]	569.8 $\pm$ 3.9	[560.5, 575.2]
MNIST	Image	10	0.01 $\pm$ 0.03	[0.00, 0.09]	96.98 $\pm$ 0.91	[94.72, 97.89]	98.76 $\pm$ 0.86	[96.64, 99.60]	98.46 $\pm$ 0.48	[97.25, 98.93]	907.8 $\pm$ 41.1	[833.4, 1006.7]
CIFAR-100	Image	100	0.62 $\pm$ 1.21	[0.00, 5.00]	82.24 $\pm$ 1.71	[74.83, 84.73]	97.06 $\pm$ 2.02	[88.32, 100.00]	89.99 $\pm$ 1.36	[83.72, 91.73]	189.3 $\pm$ 2.1	[184.8, 196.2]
FashionMNIST	Image	10	0.00 $\pm$ 0.00	[0.00, 0.00]	94.58 $\pm$ 1.02	[93.36, 97.01]	99.99 $\pm$ 0.52	[99.24, 101.04]	97.21 $\pm$ 0.53	[96.57, 98.48]	186.5 $\pm$ 1.3	[183.7, 188.0]
Speech Commands	Audio	10	1.49 $\pm$ 1.28	[0.00, 4.55]	97.59 $\pm$ 0.25	[97.15, 97.93]	100.15 $\pm$ 0.18	[99.75, 100.34]	98.04 $\pm$ 0.58	[96.60, 98.55]	459.7 $\pm$ 3.5	[454.9, 464.2]
UCF101	Video	101	3.39 $\pm$ 10.38	[0.00, 67.44]	84.65 $\pm$ 2.86	[70.53, 87.76]	96.46 $\pm$ 3.26	[80.37, 100.00]	89.84 $\pm$ 6.36	[47.19, 93.48]	117.9 $\pm$ 16.8	[89.4, 171.1]

The Grad-CAM method was also used for visual analysis [12]. In Fig. 2, it can be seen that Grad-CAM maps before and after unlearning for FashionMNIST and MNIST datasets. Before unlearning, the learned model pays attention to areas corresponding to the class of forget samples. However, after unlearning, heat maps vanish or change towards other features which are not related to forgotten sample.

V. CONCLUSION

In this paper, we proposed MIST, which is a class-level machine unlearning approach that combines sharded isolation with masked knowledge distillation. The objective of our proposed MIST is to erase the impact of the chosen forget class while keeping other class information intact with lower computational cost. As illustrated by experimental evaluations on four image datasets, the MIST framework achieves a very balanced and competitive result among forgetting efficiency, retention preservation, privacy-preserving ability, and unlearning time compared to other tested 9 methods. Moreover, class sweep experiments on image, audio, and video datasets reveal that the MIST algorithm is not confined to image unlearning and can be applied to any modality with a logit masking technique. Future work will focus on larger multimedia datasets, improved aggregation strategies, and sample-based or feature-based unlearning scenarios.

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